

1 Explainability in Reinforcement Learning

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Chapter 1

Explainability in Reinforcement Learning

Today, the majority of Artificial Intelligence (AI) agents with which humans interact are deployed in controlled or low-risk environments. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences [Silver et al., 2017]. Even more autonomous systems, such as agents based on Large Language Model (LLM), remain constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval [Chen et al., 2021, Lewis et al., 2021]. Safety-critical applications integrating machine learning components are still rare. Introducing autonomous agents in such environments, requires the establishment of strong guarantees [Seshia et al., 2022]. These requirements include in particular:

- validating the system’s functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both Reinforcement Learning (RL) and Deep Learning (DL) [Mnih et al., 2013]. As a result, the mechanisms underlying their decisions cannot be understood directly from the learned model. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, a research field has emerged: eXplainable Reinforcement Learning (XRL), a subfield of eXplainable Artificial Intelligence (XAI). The objective of XRL is to develop explainability methods that make it possible to obtain explanations of RL agents. Due to the relevance of the topic and to foster broader real-world adoption of RL, a growing number of research teams are working on developing new explainability methods. This enthusiasm is reflected in a rapid increase in the number of publications, as illustrated in Figure 1.1.¹

¹The complete search protocol, including the databases consulted and the full set of query terms, has been archived on Zenodo <https://zenodo.org/records/19387785>.

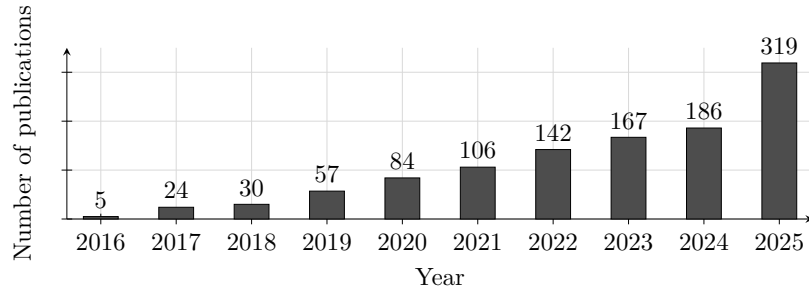


Figure 1.1: Yearly growth of Explainable Reinforcement Learning research with at least one citation on arXiv.

The growing number and diversity of these methods make it increasingly difficult to obtain a unified view of the field. This motivated our work on developing a unified taxonomy of explainability methods for RL, initially presented in [Alaarabiou et al., 2024]. The taxonomy provides a common structure for organizing existing approaches according to the type of information they use and the form of explanation they provide. Building on this work, this chapter provides a detailed presentation of the proposed taxonomy and uses it to organize a broader state-of-the-art review of XRL.

This chapter is organized as follows:

- Section 1.1 begins by clarifying what we mean by an explanation in the context of XAI.
- Section 1.1 then introduces a taxonomy for categorizing existing XRL methods.
- Section 1.3 provides a comprehensive state-of-the-art review structured around this taxonomy.
- Section 1.4 concludes the chapter with a summary and discussion.

1.1 Explanation in Artificial Intelligence

In this section, we define what an explanation is in the context of XAI and how it can be obtained. We first introduce the concept of explanation in general. We then describe how this concept applies to XAI. Finally, we further detail the properties that characterize an explainability method in the context of XAI.

1.1.1 Explanation

Establishing what is meant by an explanation is challenging due to the fundamentally multidisciplinary nature of the concept. When we refer to explanation, we are inherently referring to an explanation provided to a human. As soon as humans are involved, the problem is no longer purely computational or mathematical. It involves several fields of research, such as philosophy, psychology, and sociology. So far, these disciplines have not converged on a unified definition of explanation that can be directly applied to XAI [Miller, 2019, Bellucci et al., 2021].

For this reason, we propose the following definition of explanation. An explanation involves two entities: the object (Definition 1.1.1) being explained and the receiver of the explanation. The receiver is a human who aims to understand how the object operates and will be referred to as the observer throughout this manuscript (Definition 1.1.2). The observer seeks to understand the object and, in doing so, builds a mental model of its functioning. An explanation is therefore what allows the observer to refine and correct their mental model of how the object operates (Definition 1.1.4). This refinement is obtained through a transformation process that produces an explanation from the object being explained. We refer to this process as an explainability method (Definition 1.1.5), detailed further in Section 1.1.3. For an observer to understand an explanation, the information it contains must be interpretable (Definition 1.1.3). Information is considered interpretable when it can be directly understood by a human without requiring any additional transformation process.

Definition 1.1.1: Object

The object of an explanation is the system the observer wants to understand.

Definition 1.1.2: Observer

The observer is the human who aims to understand an object. The observer possesses a mental model of the object being explained. This mental model is an internal representation of how the observer believes the system operates. This representation may be incomplete or inaccurate.

Definition 1.1.3: Interpretable

Information is interpretable when it can be directly understood by an observer without requiring any additional transformation process.

Definition 1.1.4: Explanation

An explanation is any interpretable information that enables an observer to refine or correct their mental model of the object of the explanation.

Definition 1.1.5: Explainability Method

An explainability method is a transformation process that produces an explanation from the object being explained.

This definition of explanation provides the foundation for the approach to explainability adopted throughout this manuscript. This work adopts the perspective of *Mechanistic Hypothesis* (Hyp. 1) formalized in Hypothesis ?? [Siegel and Craver, 2024, Zednik, 2019]. Within this framework, explainability is not regarded as a binary property but as a continuum that evolves with the development of scientific knowledge.

Accordingly, the opacity of a system should be interpreted as an epistemic limitation reflecting the current state of knowledge rather than as an intrinsic property of the system itself. This view is consistent with the methodological commitments of scientific inquiry. Scientific progress is driven by the continuous

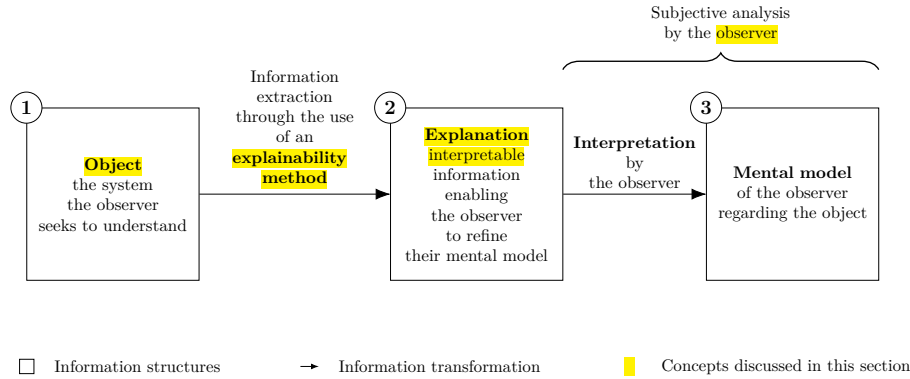


Figure 1.2: Explanation process.

91 refinement of theories and analytical methods that provide progressively deeper
 92 mechanistic accounts of observed phenomena. Consequently, explainability
 93 should be regarded as a matter of degree, increasing as these mechanisms become
 94 better understood.

Hypothesis 1: Mechanistic Hypothesis

Every system results from a sequence of causal computations. Because the rules governing these interactions are mechanical, the behavior of the system can be explained. If a system cannot be explained, this reflects a limitation of knowledge rather than an intrinsic property of the system.

95

96 This quantitative perspective allows us to express an intuitive idea: not all
 97 explanations provide the same level of value [Doshi-Velez and Kim, 2017]. The
 98 effectiveness of an explanation depends on its ability to improve the observer's
 99 understanding of the system. For example, describing a marginal aspect of
 100 a system is different from correcting an observer's misunderstanding of a key
 101 mechanism.

102 Figure 1.2 summarizes this whole explanation process, from the object of
 103 the explanation to the mental model refined by the observer, as a chain of three
 104 information structures, depicted as boxes and connected by arrows denoting a
 105 transformation of information:

- 106 (1) The **object** of the explanation, that is, the system the observer seeks to
 107 understand.
- 108 (2) The **explanation**, the interpretable information produced from the object.
 109 This transformation is performed through the use of an explainability
 110 method.
- 111 (3) The **mental model** held by the observer regarding the object.

112 A brace spanning boxes (2) and (3) indicates that this second half of the process,
 113 from the explanation to the resulting mental model, constitutes a subjective
 114 analysis carried out by the observer.

1.1.2 Explainable Artificial Intelligence

Now that the general framework of explanation has been established, we can examine how this concept applies in the context of AI. In the context of XAI, the object of explanation is an AI system. More precisely, XAI emerged as a response to the success of DL models [Gunning and Aha, 2019, Barredo Arrieta et al., 2020]. In these architectures, information is represented in a distributed manner across the network through patterns of activation and connections between computational units. Unlike symbolic approaches, individual components of Neural Network (NN) cannot be directly associated with human concepts. From this starting point, we identify two types of approaches:

- **Substitutive explainability:** In this approach, the underlying assumption is that there is nothing to explain within a NN. There is no semantic meaning associated with each computational component, and therefore nothing to interpret. This corresponds to the “black box” view of NN. Under this assumption, XAI consists of replacing as much of the DL system as possible with alternative AI methods [Rudin, 2019].
- **Intrinsic explainability:** In this approach, it is argued that NN inherently contain mechanisms that can be explained. The limitation is considered to lie not in the networks themselves, but in the absence of adequate tools to analyze their functioning [Olah et al., 2020].

This manuscript considers methods from both approaches in order to provide a comprehensive overview of the XAI field. According to the *Mechanistic Hypothesis* (Hyp. 1) adopted in this manuscript, we consider that any system can be explained. As a consequence, the perspective adopted throughout this manuscript is that of intrinsic explainability.

1.1.3 Explainability Method in XAI

Now that the general framework of XAI has been established, we need to further detail the properties of an explanation in this context. In our framework, an explanation in the field of XAI can be characterized by three components:

- **Source:** refers to the origin of the information used by the explainability method. The explainability method transforms the raw information produced by the AI system into an explanation that can be understood by an observer.
- **Form:** refers to the representation through which the explanation is communicated. The form defines how the information is presented to the observer and must allow human interpretation. An explanation can take many different forms, including text, images, computer code, and mathematical equations.
- **Subject:** refers to the aspect of the AI system that is being explained. In XAI, this distinction is commonly associated with the difference between local and global explanations. A local explanation describes why an AI system produced a specific output in a given situation. A global explanation aims to describe the general functioning of the AI system.

These three properties of an explanation are entirely determined by the

159 explainability method (Definition 1.1.5) used to produce it. In this context,
 160 an explainability method is a transformation process that maps a source of
 161 information into an explanation characterized by a specific form and subject.
 162 Research in XAI focuses on developing new methods capable of performing this
 163 transformation in order to overcome the lack of interpretability associated with
 164 DL models.

165 1.2 Taxonomy of XRL Methods

166 Now that the concept of XAI has been defined, we turn to its application in RL,
 167 known as XRL. As in XAI, explanations in XRL are produced by explainability
 168 methods and can be described in terms of their subject, source, and form. This
 169 section reviews the possible variations of these three aspects in the specific
 170 context of RL.

171 1.2.1 Subject

172 The first question to address is the subject of explanation in the context of RL.
 173 To answer this question, it is necessary to identify the different components
 174 involved in a RL system and determine which of them constitutes the object of
 175 explanation. A RL system can be described through two main components: the
 176 environment and the policy.

177 The environment defines the dynamics in which the agent evolves, including
 178 the possible states, observations, actions, and transitions. Environments are
 179 simulations designed by humans, and their dynamics are explicitly defined.
 180 Therefore, it is assumed that the environment is known to the observer and does
 181 not require explanation.

182 The main object of interest is therefore the agent's policy. The policy
 183 represents the decision-making mechanism of the agent. It is a function that maps
 184 observations to actions and is learned through interaction with the environment in
 185 order to maximize the expected cumulative reward. The policy is the component
 186 that encodes the learning process and that XRL aims to explain.

187 Explaining a policy introduces additional challenges compared with classical
 188 XAI settings. In traditional XAI problems, the objective is to explain an isolated
 189 decision, such as determining which features of an image led to a classification
 190 result. In contrast, RL introduces a temporal dimension into the decision-making
 191 process. An action is not only determined by the current observation but also
 192 influences future states and rewards. Therefore, understanding the behavior of
 193 an agent may require considering a sequence of interactions between the agent
 194 and its environment.

195 This temporal dimension leads to a distinction between two objectives of
 196 explanation in XRL:

- 197 • **Decision explanations:** focus on identifying the elements of the current
 198 observation that influenced the action selected by the policy. This type of
 199 explanation is close to classical XAI approaches, as it analyzes the rela-
 200 tionship between inputs and outputs while abstracting away the sequential
 201 nature of the decision process.
- 202 • **Strategy explanations:** aim to characterize the agent’s behavior over
 203 time. They focus on sequences of decisions and interactions with the
 204 environment in order to understand how the agent achieves its long-term
 205 objective. This type of explanation is specific to RL, as it addresses the
 206 temporal and goal-oriented nature of the learning process.

207 These two objectives should not be considered as strictly separated categories,
 208 but rather as two extremes of a continuum. Understanding the elements of an
 209 observation that influence a specific decision can help the observer generalize
 210 this information to infer broader aspects of the agent’s behavior. Conversely,
 211 understanding the agent’s overall strategy can provide insights into the reasoning
 212 behind individual decisions.

213 1.2.2 Source

214 We identify three sources of explanation in RL, all obtained during or after the
 215 learning process and containing information about the agent:

- 216 • **Policies:** policies are functions that generate a distribution over the action
 217 space for a given observation. The objective of RL is to learn a policy that
 218 maximizes the expected cumulative future rewards. During the learning
 219 phase, the policy is iteratively improved through interactions with the
 220 environment.
- 221 • **Trajectories:** trajectories represent sequences of successive interactions
 222 between the agent and the environment. At each step, the agent selects
 223 an action based on its observation, which modifies the environment and
 224 produces a reward as well as a new observation. This process continues
 225 until the end of the trajectory.
- 226 • **Value functions:** value functions provide predictions about the future
 227 outcome of a trajectory from a given observation. The most commonly
 228 used value function is the critic function, which estimates the expected
 229 sum of future rewards. This prediction guides the learning process, as the
 230 agent updates its policy to maximize the estimated future return. However,
 231 other types of value functions can also be used as sources of explainability.

Explanation Subject	Explanation Form	Explainability Method Family	Source of Explanation		
			Policy	Trajectories	Value functions
Decision	Policy Model	<i>Interpretable policy learning</i>	✓		
		<i>Policy approximation via interpretable model</i>	✓		
	Saliency Map	<i>Observation perturbation</i>	✓		
		<i>Observation gradient analysis</i>	✓		
		<i>Attention-based architecture</i>	✓		
	Counterfactual	<i>Latent space usage for counterfactual generation</i>	✓		
Strategy	Agent Trajectory	<i>Hierarchical decomposition</i>	✓	✓	
		<i>Extraction via critic analysis</i>		✓	✓
	Trajectory Prediction	<i>Critic enrichment</i>			✓
	Interaction Model	<i>Bayesian network learning</i>		✓	
		<i>Latent space usage for MDP generation</i>	✓		

Table 1.1: Taxonomy of explainability methods in reinforcement learning.

1.2.3 Form

We identify six forms of explanation in the RL setting:

- **Policy Model:** an interpretable model of the agent’s policy function.
- **Saliency Map:** a weighting of the input features representing their influence on the output of the policy function.
- **Counterfactual:** a modified observation obtained by perturbing the original observation in order to produce a different action from the agent.
- **Agent Trajectory:** a sequence of successive interactions between the agent and the environment.
- **Trajectory Prediction:** a prediction about the future evolution of the trajectory produced by a value function.
- **Interaction Model:** an interpretable model of the interactions between the agent and the environment.

Among the three components characterizing an explanation, the form is the one that varies most significantly across explainability methods. While subjects and sources tend to be relatively stable in the XRL literature, the diversity in how explanations are presented is considerably greater. The list above is by no means definitive and may evolve as new approaches emerge. For this reason, the following section is organized according to the form of explanations and presents existing explainability methods based on this criterion.

1.3 State-of-the-Art Review of XRL Methods

The purpose of this section is to detail the functioning of existing explainability methods (see Table 1.1). They are presented according to the form of explanation they produce, following the taxonomy introduced in Section 1.2.

Each subsection introduces one of the six forms of explanation identified in Section 1.2.3. Within each subsection, each subsubsection is dedicated to a family of methods capable of producing that form of explanation.

A family of methods is defined as a set of methods that share the same source, form, and subject of explanation, as well as a similar procedure for generating explanations. Methods belonging to the same family may differ in their implementation or in the details of the generation process, while relying on the same underlying principle.

1.3.1 Policy Model

The policy function can be represented by a wide variety of function classes. In modern RL, NN is the most commonly used models for representing policies. Although these models achieve the highest performance, their connectionist nature makes their internal representations non-interpretable.

For this reason, a substitutive approach to XRL aims to replace these models with alternative models considered interpretable. Such models can be understood through direct analysis of their representations. The main classes of interpretable models used to represent policy functions include decision trees [Topin et al., 2021], algorithms [Trivedi et al., 2022], formal logic [Payani and Fekri, 2019], and linear models [Garreau and von Luxburg, 2020].

Explanations derived from an interpretable model primarily provide insights into the decision-making process, since what is modeled is the mapping between an observation and an action. If the model is sufficiently simple or well structured, the observer can form a mental representation of a sequence of decisions across the environment and thus obtain information about the agent’s strategy.

The main limitation of these methods is that interpretable models rely on symbolic representations. When the policy to be represented is too complex, achieving comparable performance requires a large number of rules or conditions. The resulting model can quickly become difficult to analyze and lose its interpretability.

For example, representing a policy capable of playing Go using a symbolic model would require considering a very large number of possible situations and specific cases. Even if such a representation were possible, the resulting model would likely become intractable for human analysis due to its complexity. This illustrates why NN-based approaches have become dominant in such domains, as they can learn complex decision strategies without requiring an explicit enumeration of rules.

To obtain an interpretable policy model, two families of explainability methods are available: *interpretable policy learning* and *policy approximation via interpretable model*.

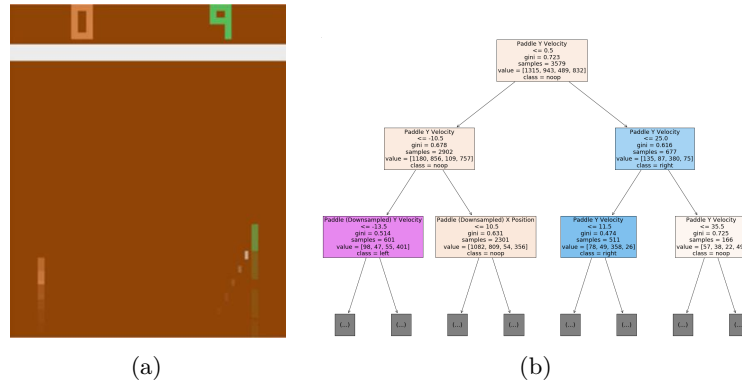


Figure 1.3: Policy approximation via an interpretable model [Sieusahai and Guzdial, 2021]. (a) shows an observation from the Atari game “Pong”, and (b) shows the decision tree obtained by approximating a deep reinforcement learning policy trained on this environment.

Interpretable Policy Learning

This family of methods aims at learning a policy represented by an **interpretable** model [Topin et al., 2021, Trivedi et al., 2022]. It is the most direct approach to addressing the challenges posed by XRL. This type of training requires modifying the Markov Decision Processes (MDP) in order to adapt it to the search space and to the type of policy representation. Moreover, this search space often needs to be reduced to make learning feasible. Such a **reduction** requires the integration of **expert knowledge** into the learning process. With this family of methods, the *interpretable policy function* itself is used to provide explanations to the observer.

Policy Approximation via Interpretable Model

An interpretable model can be trained to approximate a policy learned through deep RL [Sieusahai and Guzdial, 2021]. In this setting, the problem is formulated as a supervised learning task, where the original deep RL policy is used to label each observation with the action it selects (see Figure 1.3a). Using this dataset, the interpretable model is trained to reproduce the behavior of the original policy, yielding for instance the decision tree illustrated in Figure 1.3b. The resulting interpretable model is then used as an explanation for the observer. If this approximation is sufficiently faithful, it can also be deployed in the environment in place of the original policy, thereby providing a high degree of control over the agent.

1.3.2 Saliency Map

The influence of each part of the input on the action selected by the policy can be visualized using heatmaps (Figure 1.4). In the RL setting, the input corresponds to the observation received by the agent at a given time step. Heatmaps therefore highlight the regions of the observation that have the greatest influence on the action selection process. By providing this information, they allow the observer

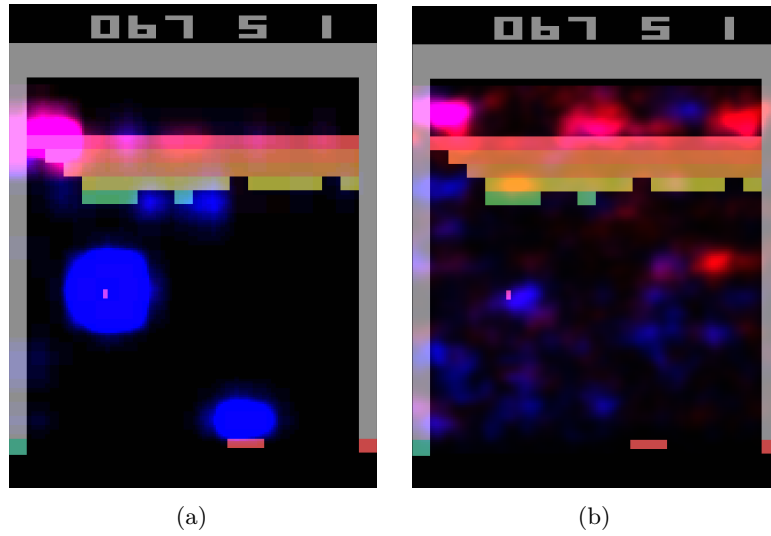


Figure 1.4: Heatmaps obtained by [Greydanus et al., 2018] for the same observation of the Atari game “Breakout”. Highlighted areas correspond to the most influential regions for the policy function’s decision. (a) is obtained via observation perturbation (see Section 1.3.2), whereas (b) is obtained via observation gradient analysis (see Section 1.3.2).

to identify which elements of the environment are considered important by the policy when making a decision.

A limitation of heatmap-based methods is that the resulting explanations are not always directly meaningful to the observer. As illustrated in Figure 1.4b, interpreting which regions are important often requires human reasoning and involves a degree of subjectivity. This limitation becomes particularly relevant in RL, where the objective is not only to explain a single decision but also to understand the factors influencing a sequence of decisions over time.

Three families of explainability methods can be used to obtain this type of explanation: *observation perturbation*, *observation gradient analysis*, and *attention-based architectures*.

Observation Perturbation

Perturbation analysis consists in applying various **modifications** to the observation in order to study their **impact** on the output of the **policy function** [Greydanus et al., 2018]. The greater the variation in the output caused by a perturbation applied to a given component of the input vector, the higher the **weight** assigned to that component. The magnitude of the perturbation is measured by computing the distance between the original output and the output obtained from the perturbed input. Using this set of weights, a *heatmap* is constructed, enabling the observer to visualize which parts of the observation vector are most sensitive in the action-selection process (see Figure 1.4a).

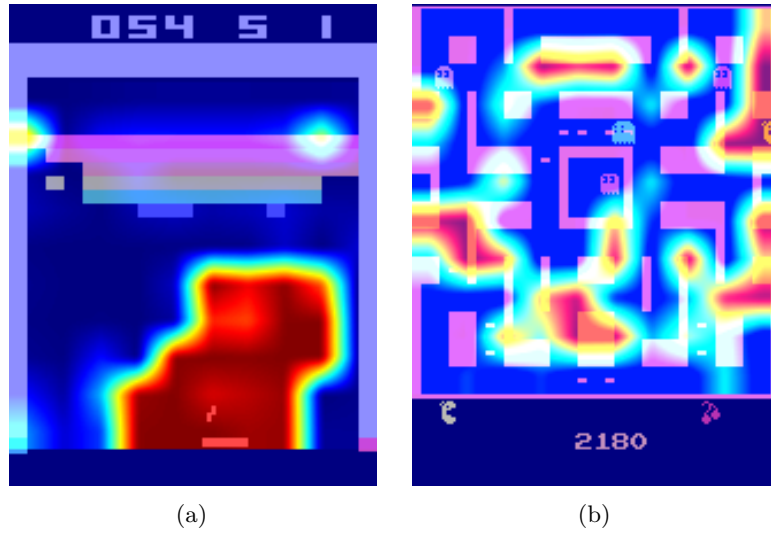


Figure 1.5: Attention-based heatmaps obtained by [Mott et al., 2019] for the Atari games “Breakout” and “Ms. Pac-Man” (see Section 1.3.2). In (a), the highlighted regions are easily interpretable, whereas in (b), the attention is spread across many regions of the observation, making the explanation much harder to interpret.

343 *Observation Gradient Analysis*

344 Observation **gradient** analysis refers to a set of methods that require a differen-
 345 tiable policy function in order to extract information from it [Simonyan et al., 2014,
 346 Weitkamp et al., 2019, Huber et al., 2019]. To this end, the gradient of the pol-
 347 icy function is computed with respect to all **components of the input vector**.
 348 For each input component, the resulting gradient provides a weighting that
 349 represents its importance in the output of the policy function. Based on these
 350 weightings, a *heatmap* is constructed to allow the observer to visually identify
 351 the parts of the input vector that are the most influential in the policy function’s
 352 output (see Figure 1.4b).

353 *Attention-Based Architecture*

354 **Attention** blocks constitute a NN architecture that makes it possible to weight
 355 the relative importance of one part of a vector with respect to the other parts
 356 of the same vector [Vaswani et al., 2023]. These weights are computed as a
 357 function of the vector itself and are learned by the NN during training. For
 358 a given observation, each component of the observation vector is assigned a
 359 scalar value. It can therefore be assumed that the components associated
 360 with the highest weights are the most influential in determining the output
 361 [Mott et al., 2019, Itaya et al., 2021]. This attention-based *heatmap* is then
 362 provided to the observer as the explanation (see Figure 1.5a). However, as
 363 shown in Figure 1.5b, the attention weights are not always concentrated on a
 364 small, easily interpretable set of regions, which can make this type of explanation
 365 difficult to exploit.

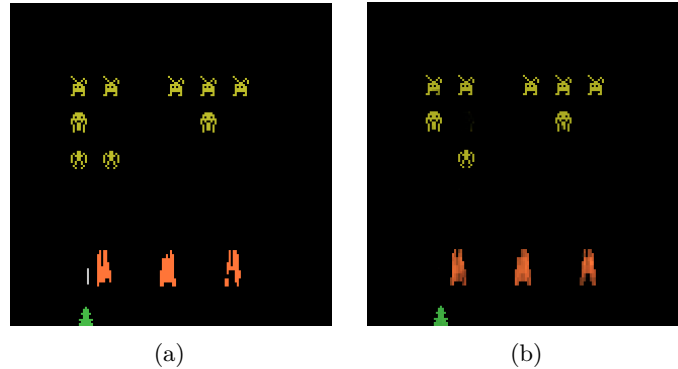


Figure 1.6: Counterfactual explanation of the agent’s behavior for the game *Space Invaders* [Olson et al., 2021]. (a) is the original observation, for which the agent selects the action “LEFT”, whereas (b) is the counterfactual observation, for which the agent selects the action “RIGHT_FIRE”.

1.3.3 Counterfactual

In the context of RL, counterfactual explanations are used to generate new observations. This new observation, distinct from the original one, leads the agent to adopt an alternative behavior. A counterfactual explanation aims to explain the agent’s decision-making process. Modifying the observation informs the observer about the adjustments required for the agent to adopt a different behavior. The family of methods available to generate counterfactuals is the *use of the Latent Space (LS) for counterfactual generation*.

Latent Space Usage for Counterfactual Generation

In this family of methods, **generative models** are used [Bond-Taylor et al., 2021] to exploit the Latent Representation (LR). This type of model learns to generate new data by imitating the training data distribution. In this context, generative models aim to produce alternative observations that are likely to alter the agent’s actions (see Figure 1.6) [Olson et al., 2021].

The main challenge of this approach lies in the fact that, to serve as explanations, counterfactuals must be both **close** and **feasible**. A close counterfactual refers to one that does not differ significantly from the original observation. A feasible counterfactual corresponds to an observation that is meaningful within the considered environment.

1.3.4 Agent Trajectory

Until now, we have presented methods that focus on explaining individual decisions made by the agent. We now consider methods that aim to provide insight into the agent’s strategy by analyzing its behavior over time.

The most direct approach consists in visualizing agent trajectories, most commonly through complete episodes. By analyzing these trajectories, the observer can infer the dynamics underlying the agent’s behavior and how its decisions evolve through time. This type of visualization is not specific to XRL.

In practice, visualizing agent episodes is a standard step in the development and evaluation of RL. It provides a simple way to verify that the agent behaves as expected and to detect potential implementation errors.

This simple approach can be further enhanced using families of methods such as *hierarchical decomposition* and *extraction via critic analysis*.

Extraction via Critic Analysis

In this family of methods, the predictions of the critic are used to determine which **trajectories to present to the observer** [Sequeira and Gervasio, 2020]. A set of states is selected from multiple trajectories. For a given state, all possible actions are examined. The state is assigned a value equal to the difference between the critic’s prediction for the lowest-valued action and the prediction for the highest-valued action. Trajectories containing the states with the highest values according to this procedure are then chosen to be presented to the observer. These trajectories are considered **critical moments** of the agent and are therefore relevant for analysis. For this family of methods, the *selected trajectories* represent the interpretable information provided to the observer.

Hierarchical Decomposition

Hierarchical agents are composed of a set of **sub-policies** specialized for specific tasks within subsets of the environment’s states. This family of methods distributes the optimization of the reward function across multiple models, significantly simplifying the learning process. Moreover, this **decomposition** of the policy function can also provide explainability in the agent’s strategy [Beyret et al., 2019]. If a sub-policy is selected, it indicates that the agent will execute a specialized behavior. This behavioral specialization provides insight into the strategy the agent employs to maximize the reward function. The agent’s *trajectory*, together with the *sub-policy* selected for each action, is used as interpretable information.

1.3.5 Trajectory Prediction

Making predictions about the future of a trajectory allows one to form a representation of what the system expects, based on its training experience. These predictions are produced using value functions. Such functions provide insights into the future evolution of the agent’s trajectory. By their nature, this form of explanation informs the observer about the strategy adopted by the agent. Value functions provide an estimate of the expected cumulative reward from a given state. Consequently, a value function maps an observation to a scalar value in $\mathbb{R}$.

This aggregated prediction provides limited information about the underlying factors that contribute to this value. It does not directly reveal which aspects of the task are being optimized or how the predicted return is composed. To improve the interpretability of value-based explanations, a family of methods known as *critic enrichment* can be employed.

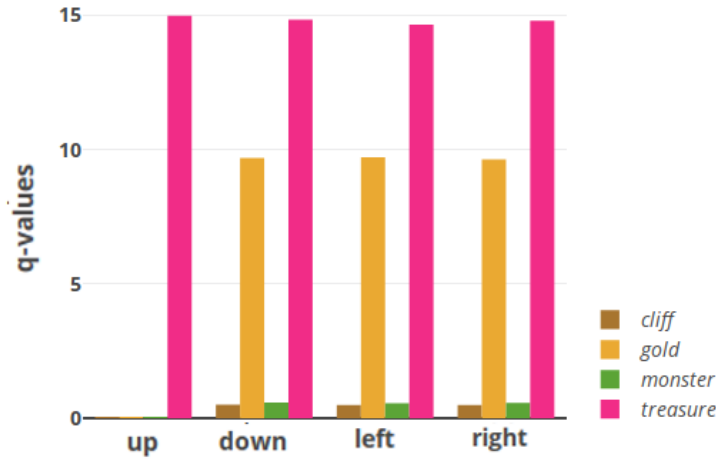


Figure 1.7: Prediction of sub-rewards as a function of the action selected by the agent for a given state [Juozapaitis et al., 2019]. Depending on the action chosen by the agent, one can determine which sub-reward it aims to maximize.

Critic Enrichment

In order to obtain interpretable information from the predictions of the **critic**, it is possible to **decompose** the critic into multiple **sub-functions** [Juozapaitis et al., 2019]. The reward function is thus decomposed into multiple sub-reward functions, each representing a **sub-objective** of the task to be accomplished. Similarly, for each of these sub-reward functions, corresponding sub-critic functions are defined, along with a function that combines their outputs. During training, the agent selects its actions so as to maximize this combination function of the sub-critics. Using this approach, for each observation and action, a prediction is obtained regarding the sub-objectives the agent is attempting to maximize (see Figure 1.7). This *prediction* is then used as the explanation for the observer.

1.3.6 Interaction Model

Modeling the interactions between an agent and its environment makes it possible to understand how the agent's decisions are made and what impact they have on the environment. By incorporating these interactions into an explanatory model, one can better grasp the complex relationships between the agent and its environment, thereby identifying the determining factors in the decision-making process. By explicitly handling the agent–environment interaction, this type of explanation aims to explain the agent's strategy. The families of methods used to obtain explanations in the form of agent–environment interactions are: *Bayesian network learning* and *LS usage for MDP generation*.

Bayesian Network Learning

The evolution of an agent within an environment can be observed as a set of **random variables** connected through a **Bayesian network** [Madumal et al., 2020]. Random variables are defined to represent both external aspects (the environ-

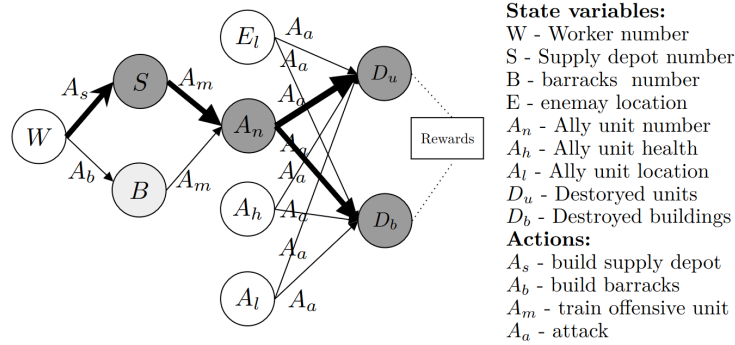


Figure 1.8: Agent-environment interaction model for the video game *StarCraft II* [Madumal et al., 2020]. Nodes represent random variables, and edges represent causal relationships between them.

ment) and internal aspects of the agent (the actions). By analyzing trajectories, it becomes possible to infer causal relationships among these random variables. At the end of this process, a weighted causal graph is obtained, which serves as an explanation for the observer (see Figure 1.8). This *Bayesian network*, representing the interactions between the agent and the environment, is then used as the explanation.

Latent Space Usage for MDP Generation

Each layer of a NN can be viewed as a projection from one space to another [Zahavy et al., 2017]. As information propagates through successive layers of a NN, it becomes increasingly abstract. It can therefore be assumed that two vectors that are close in this projected space are also close in terms of the agent's perception. Based on this observation, it is possible to **redefine an MDP** by leveraging the **abstract representations** produced by the final layers of a NN. Once the new MDP has been constructed, a model of the interaction between the agent and its environment is obtained. This model, represented as a *graph*, is then used as the explanation.

Figure 1.9 illustrates this idea using a two-dimensional *t-SNE* projection of the LS of the NN used to represent the policy. Each point is colored according to the value function's prediction for the corresponding state. An agreement can be observed between the structure of this projected space and the critic's predictions: states that are assigned similar values by the critic also tend to be located close to one another in the projection. This agreement makes it possible to manually identify clusters of states, which likely correspond to states that are perceived as similar by the agent.

1.4 Conclusion

The taxonomy and the state-of-the-art review presented in this chapter make it possible to draw three main observations.

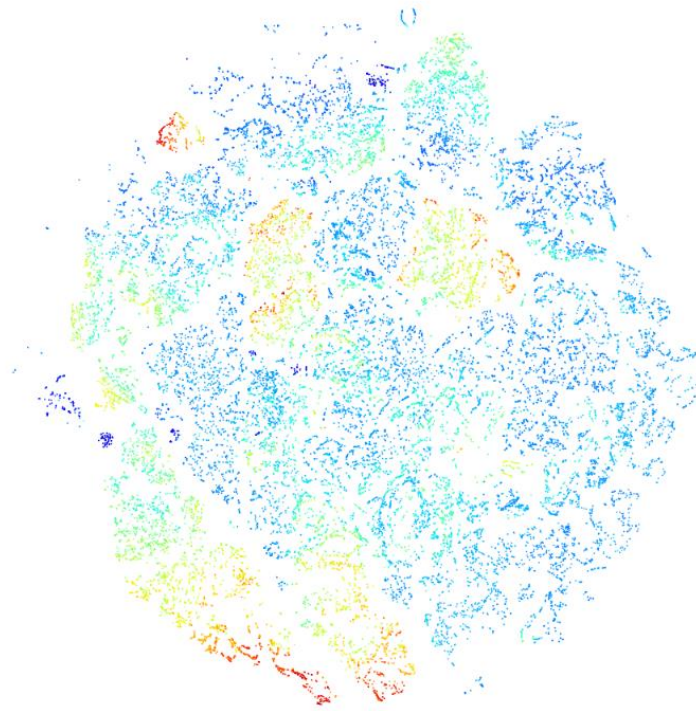


Figure 1.9: t -SNE projection of a neural network's latent space [Zahavy et al., 2017]. Each point corresponds to a state, and its color represents the value function's prediction for that state, with warmer colors indicating higher predicted values.

486 1.4.1 Absence of Explanation of the Learning Process

487 Regarding the subject of explanation, Table 1.1 shows that none of the reviewed
 488 methods addresses the learning process itself. Such an explanation would concern
 489 how the policy is progressively updated throughout training, and why it evolves
 490 over successive iterations to converge toward a particular solution. Several
 491 reasons explain this absence.

492 Among the different machine learning paradigms, RL is arguably the one that
 493 would benefit the most from explanations of the learning process. Unlike super-
 494 vised or unsupervised learning, it relies on an explicit exploration–exploitation
 495 trade-off, making the learning trajectory itself an informative object of explana-
 496 tion. Most XRL methods, however, are inherited from the XAI literature, where
 497 explanations focus almost exclusively on the behavior of already-trained models.
 498 This limitation has naturally carried over to XRL.

499 From a practical perspective, explaining the learning process is also of limited
 500 industrial interest. In most real-world applications, practitioners deploy an
 501 already-trained agent, while training itself takes place offline. Users are generally
 502 interested in understanding the behavior of the final policy rather than the
 503 optimization process that produced it.

504 Explaining the learning process also raises significant methodological chal-

505 lenges. Training in RL can be viewed as the iterative improvement of a policy
 506 through repeated interactions with the environment. In this sense, explaining
 507 learning would amount to explaining the evolution of an entire sequence of
 508 policies rather than a single one. Current XRL methods still struggle to provide
 509 comprehensive explanations of an individual policy, so extending them to a
 510 full training trajectory appears considerably more difficult. Explanations of
 511 the learning process therefore remain an open research direction rather than a
 512 straightforward extension of existing XRL methods.

513 1.4.2 Lack of Unified Metrics for Explainability

514 The diversity of explanation forms identified in this review also reveals another
 515 limitation shared by nearly all families of XRL methods, which is the absence
 516 of reliable metrics for evaluating explainability. In many studies, explanations
 517 are not evaluated with human participants. Instead, authors focus on technical
 518 properties of the proposed method, such as fidelity, sparsity, or computational
 519 efficiency. When human evaluations are conducted, they usually rely on subjective
 520 satisfaction questionnaires. Although useful, self-reported satisfaction remains
 521 prone to numerous biases and does not necessarily indicate that the explanation
 522 has improved the observer’s mental model or understanding of the agent.

523 This limitation is not specific to XRL, it reflects a broader challenge affecting
 524 XAI as a whole. In our view, this challenge stems from a historical development
 525 largely driven by the computer science community. Explainability has conse-
 526 quently often been studied from an algorithmic or mathematical perspective,
 527 with less attention given to how humans actually benefit from explanations.

528 Our definition of explanation explicitly places the observer at the center of
 529 the explanatory process. From this perspective, the value of an explanation
 530 should not only depend on its technical properties but also on its ability to
 531 improve the observer’s understanding. However, the field still lacks objective
 532 and standardized metrics capable of measuring this aspect.

533 Addressing this challenge will require stronger interdisciplinary collaboration
 534 between computer science and disciplines such as cognitive science, psychology,
 535 philosophy, and sociology. These fields provide the theoretical and experimental
 536 tools needed to study human understanding and to develop evaluation protocols
 537 that measure the actual effectiveness of explanations. The absence of such
 538 collaborations is likely a consequence of the relative youth of XAI as a research
 539 field, whose theoretical foundations and evaluation methodologies are still being
 540 established.

541 1.4.3 Toward Intrinsic Explainability in XRL

542 Our review also suggests that current XRL research remains largely dominated
 543 by substitutive explainability approaches (Section 1.1). Many existing methods
 544 explain RL agents by constructing surrogate models. While these approaches
 545 can provide useful explanations for simple policies, their ability to scale to
 546 increasingly complex agents is limited. This limitation is particularly important
 547 in deep RL, where NN are used to represent complex policies that would be
 548 difficult to approximate with simpler surrogate models. Replacing such a model
 549 with a simpler one therefore risks losing precisely the complexity that makes
 550 deep RL effective.

551 For this reason, we argue that a promising direction for XRL is to develop
552 explanations directly from the NN underlying the agent. Rather than bypassing
553 the model, such approaches aim to analyze the representations learned by the
554 network itself. From this perspective, the NN should not necessarily be regarded
555 as an inherently opaque black box, but as a complex representation whose
556 apparent opacity may instead reflect the limitations of current analysis tools.

557 Among the methods reviewed, *latent space usage for MDP generation* (Sec-
558 tion 1.3.6) appears closest to this intrinsic perspective. Instead of replacing
559 the learned representation, it directly exploits the LS to identify meaningful
560 concepts that can support the explanation of the agent’s behavior. However, this
561 approach still relies on a largely manual and qualitative analysis of the LS, which
562 limits its reproducibility and its ability to produce systematic explanations. The
563 next chapter builds upon this idea and proposes a mechanistic XRL approach
564 based on LS analysis.

565 Acronyms

566 **AI** Artificial Intelligence 1, 5

567 **DL** Deep Learning 1, 5, 6

568 **LLM** Large Language Model 1

569 **LR** Latent Representation 13

570 **LS** Latent Space 13, 15, 16, 19

571 **MDP** Markov Decision Processes 10, 15, 16

572 **NN** Neural Network 5, 9, 12, 16, 18, 19

573 **RL** Reinforcement Learning 1, 2, 6–11, 13, 14, 17, 18

574 **XAI** eXplainable Artificial Intelligence 1, 2, 5–7, 18

575 **XRL** eXplainable Reinforcement Learning 1, 2, 6–10, 13, 17–19

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